



Global Public Health Data Visualization

Project Report

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Problem and Stakeholders

This project was created to solve a common problem for students in public health: real-world datasets are often messy and hard to read. It takes time to repeatedly filter data and recreate plots in R. Because of this, important patterns about where and when an outbreak occurred are not easily visible. The dataset used was `country_timeseries.csv`, which tracks the 2014-2015 Ebola outbreak. In its raw form, the dataset has too many columns and is not suitable for quick communication with students.

Stakeholders involved:

- **Students and instructors in public health:** They need an interactive tool to explore the dataset and understand how an outbreak spreads over time.
- **Non-technical audiences:** They benefit from simple, readable visual summaries instead of raw tables.

Why I did this project The main problem was the lack of an easy way to explore which countries were most affected by Ebola without rewriting code. I wanted to create a simple interactive application that allows users to filter by country, time period, and case type (Cases vs Deaths) and immediately see how the visual summaries change.

Result of the project The final result is a simple, two-tab Shiny dashboard that allows users to dynamically explore the trend of the outbreak over time and see the total impact by country. This tool makes the dataset more understandable and suitable for teaching and basic exploration.

Pathway to the Solution To solve this problem, I designed an interactive Shiny application in R that transforms a raw CSV dataset into a simple visualization tool.

- **Step 1:** Selecting variables and defining goals: I selected the top 5 affected countries (Liberia, Sierra Leone, Guinea, Nigeria, Mali), date timelines, and case types (Cases or Deaths).
- **Step 2:** Preparing the data: Before visualization, I cleaned the data. I converted date strings into simple Date formats and used basic reshaping to put Cases and Deaths into readable columns. Missing values were removed to keep charts clean.
- **Step 3:** Designing the interactive interface: I created three simple filters: a dropdown for the Country, a dropdown for the Outbreak Stage, and radio buttons for Cases or Deaths.
- **Step 4:** Choosing visualizations: I included two complementary plots organized in tabs: a Line Plot to show the trend over time, and a Bar Chart to show the total counts per country.
- **Step 5:** Ensuring robustness and clarity: I added a control that displays a clear message ("No data for the selected filters") when filtering removes all data for a specific subgroup, avoiding empty or broken plots.

Methods and Technical Approach

- **Tools used:** R Shiny (for the app), ggplot2 (for plots), dplyr& tidyr (for data cleaning).

- **Data preparation:** Load the CSV, convert dates, reshape the columns, and use `drop_na()`.
- **User interface design:** A simple sidebar with dropdowns and a main panel with two tabs.
- **Server logic:** A single reactive dataset filters the data based on simple if statements for the dates and countries, updating both plots automatically.

Reproducibility and sharing The code includes automatic installation of required packages. The application can be run locally by anyone with RStudio.

What problem was solved and what was achieved This project aimed to make a historical public health dataset easier to understand for students. By building this interactive dashboard, I transformed a raw dataset into a clear and user-friendly visualization tool. It helps users better understand the scale of the Ebola outbreak and compare countries without advanced programming.

Github: <https://github.com/DornaheidaryGPH/Ebola-Data-Visualization.git>

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